

Speech Emotion Recognition using Deep learning by hybridizing CNN-BiLSTM and attention mechanism

Paper ID: 2555
Track No. : 3

Muni Rangadu K, Reddi Sekhar M, Anjali K, Ngangbam Indrason, School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India

Presented by: Muni Rangadu K, School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh, India

Speech Emotion Recognition (SER)

What it is: Technology that identifies human emotions from the way people speak.

Speech Emotion Recognition (SER) is crucial for enhancing human-computer interaction. Its applications span multiple fields such as healthcare, education, and customer service. By accurately detecting emotions, SER can improve communication and understanding, addressing existing challenges like noise and limited emotional range.



Literature Survey

Classical ML: SVMs, GMMs, HMMs with handcrafted acoustic features.

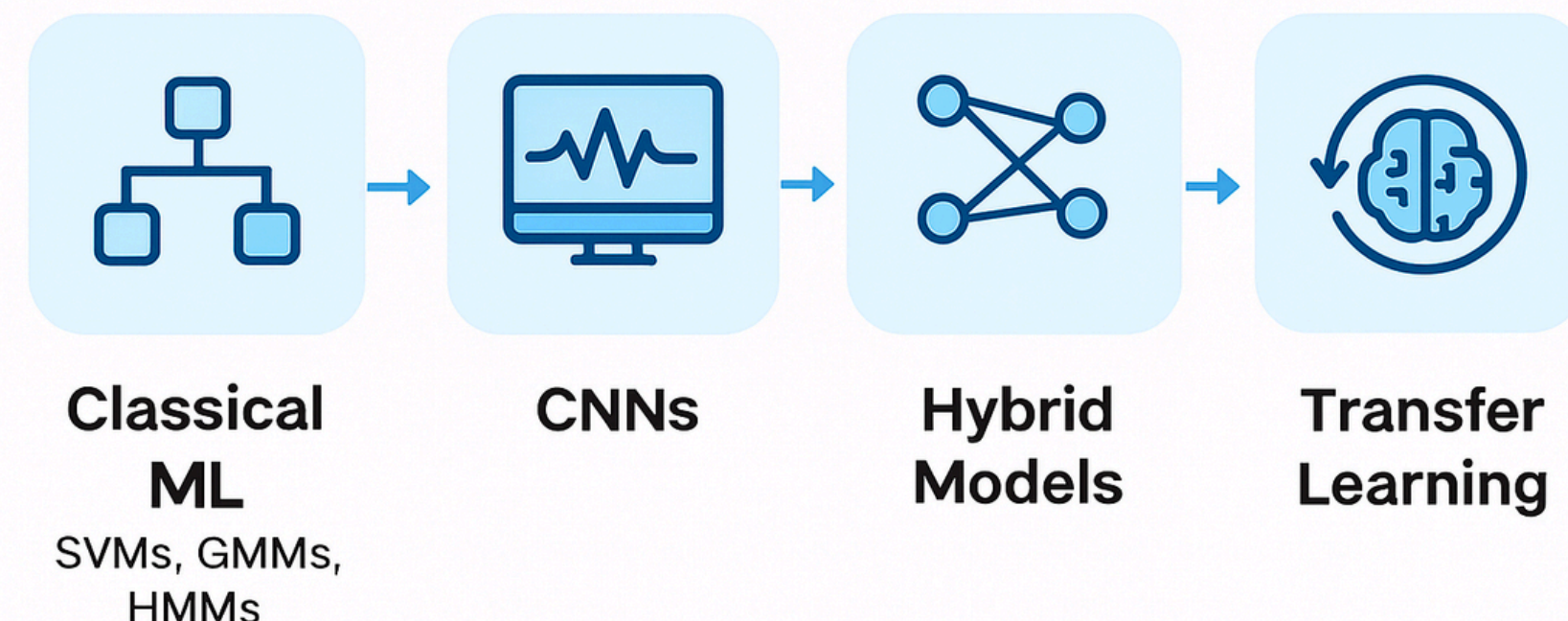
CNNs: Learned spectrogram-based patterns, improving robustness.

RNNs/LSTMs: Captured temporal dynamics but lacked fine detail.

Hybrid Models + Attention: Combined CNN–BiLSTM for better emotional focus.

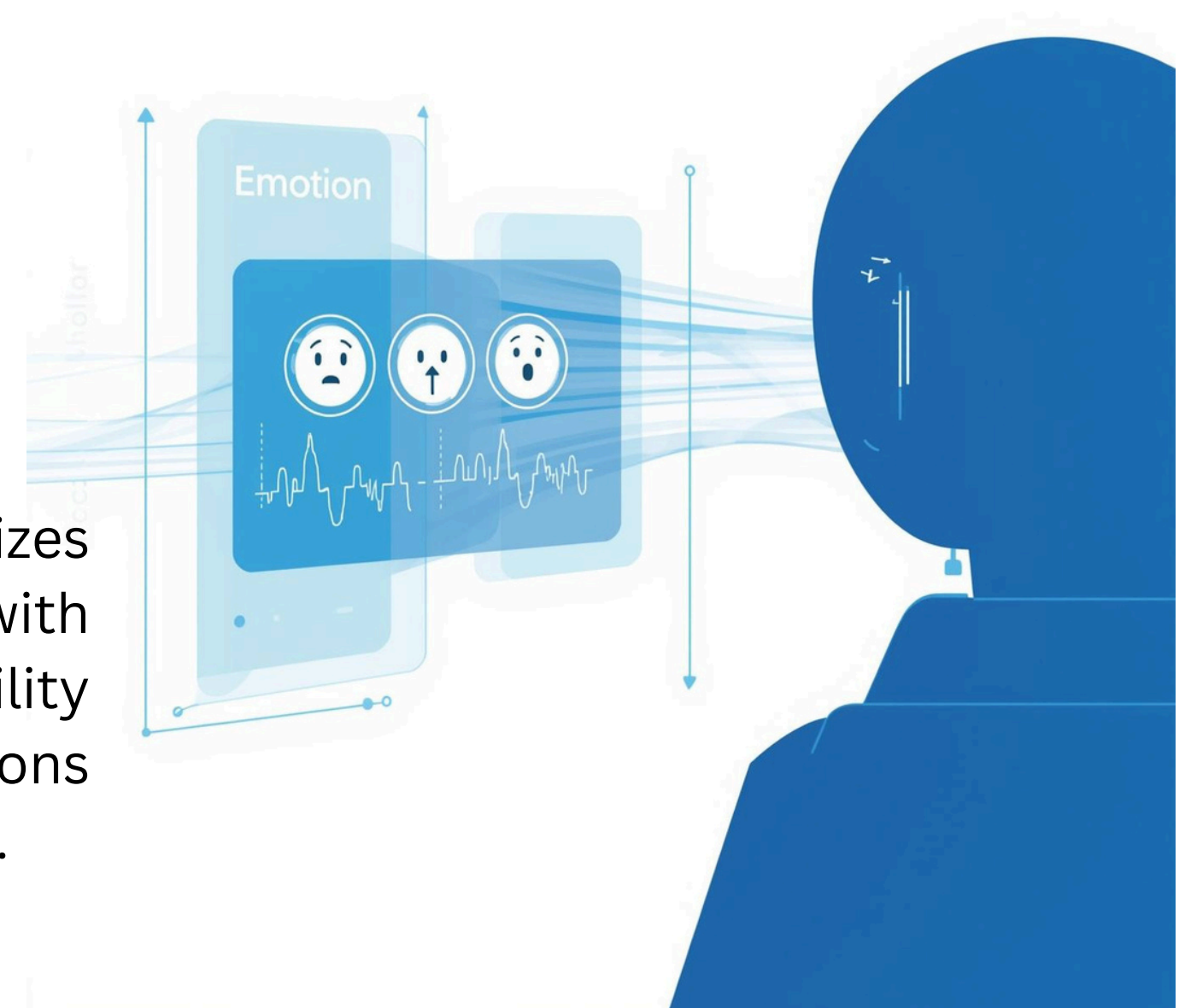
Transfer Learning: Leveraged wav2vec, HuBERT for low-resource and real-world SER.

Evolution of Speech Emotion Recognition



Methodology

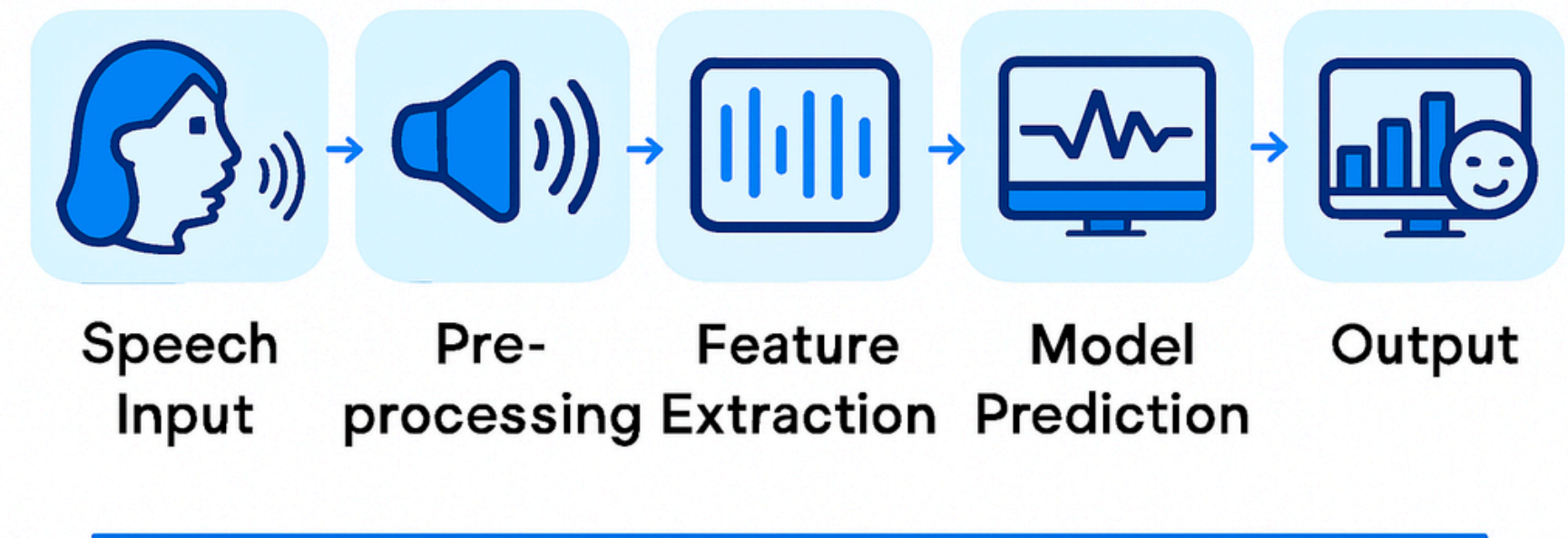
The growing demand for real-time emotion recognition emphasizes the necessity of deploying systems that can seamlessly integrate with graphical user interfaces and transcription services. This capability enhances user experience and allows for more effective applications across various domains, including healthcare and customer service.



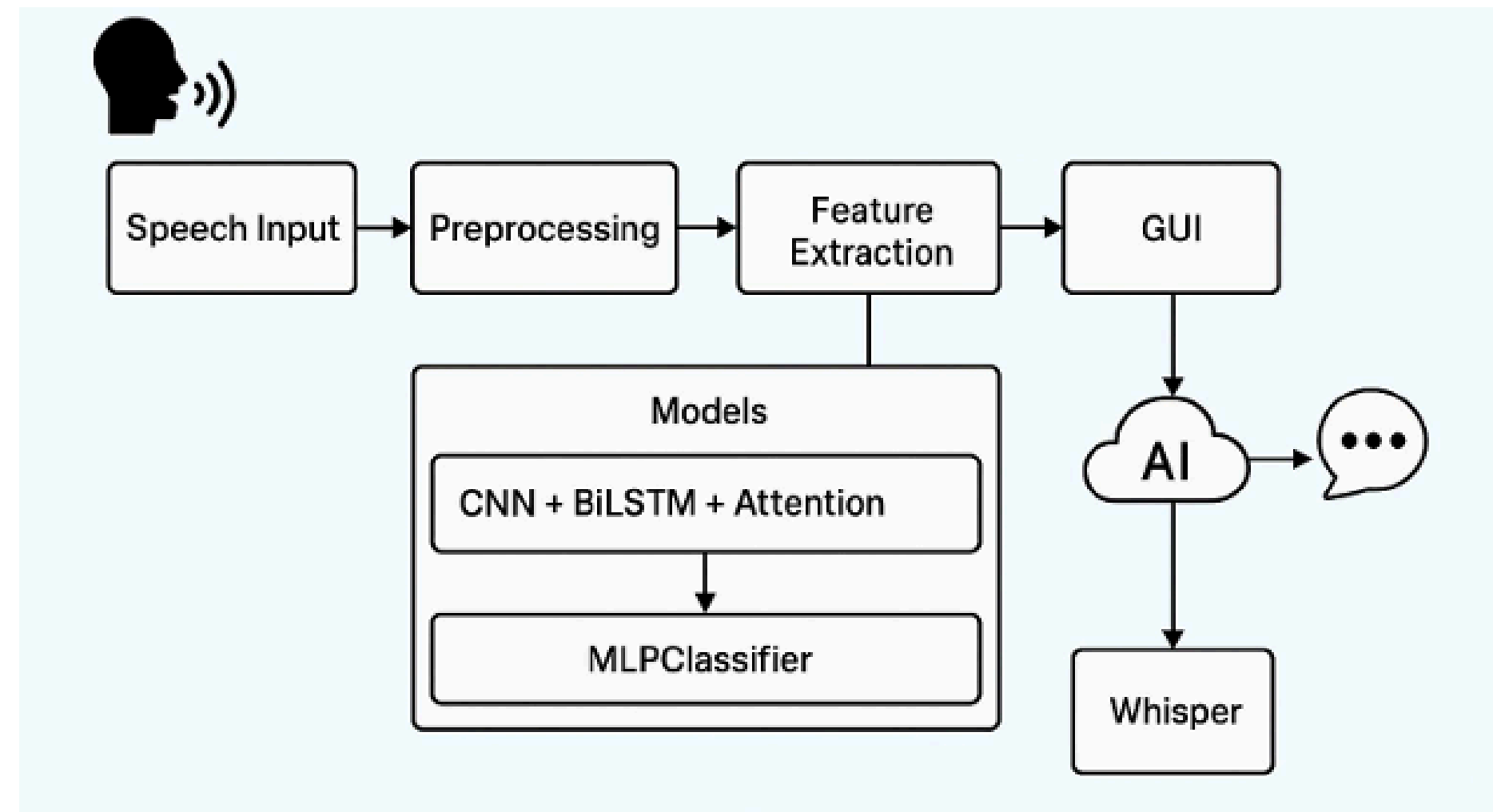
Architecture

This architecture outlines the workflow for Speech Emotion Recognition, comprising steps from speech input to preprocessing, feature extraction, through the CNN and BiLSTM layers, culminating in the application of attention mechanisms and integration with a GUI and Whisper transcription for real-time results.

Proposed Architecture



CNN–BiLSTM Attention



Block diagram

This section explores the integral roles of the Convolutional Neural Network (CNN) for feature extraction, the Bi-Directional Long Short-Term Memory (BiLSTM) for sequence processing, and the Attention mechanism to enhance the model's focus on relevant inputs for improved speech emotion recognition.

Experimental Setup

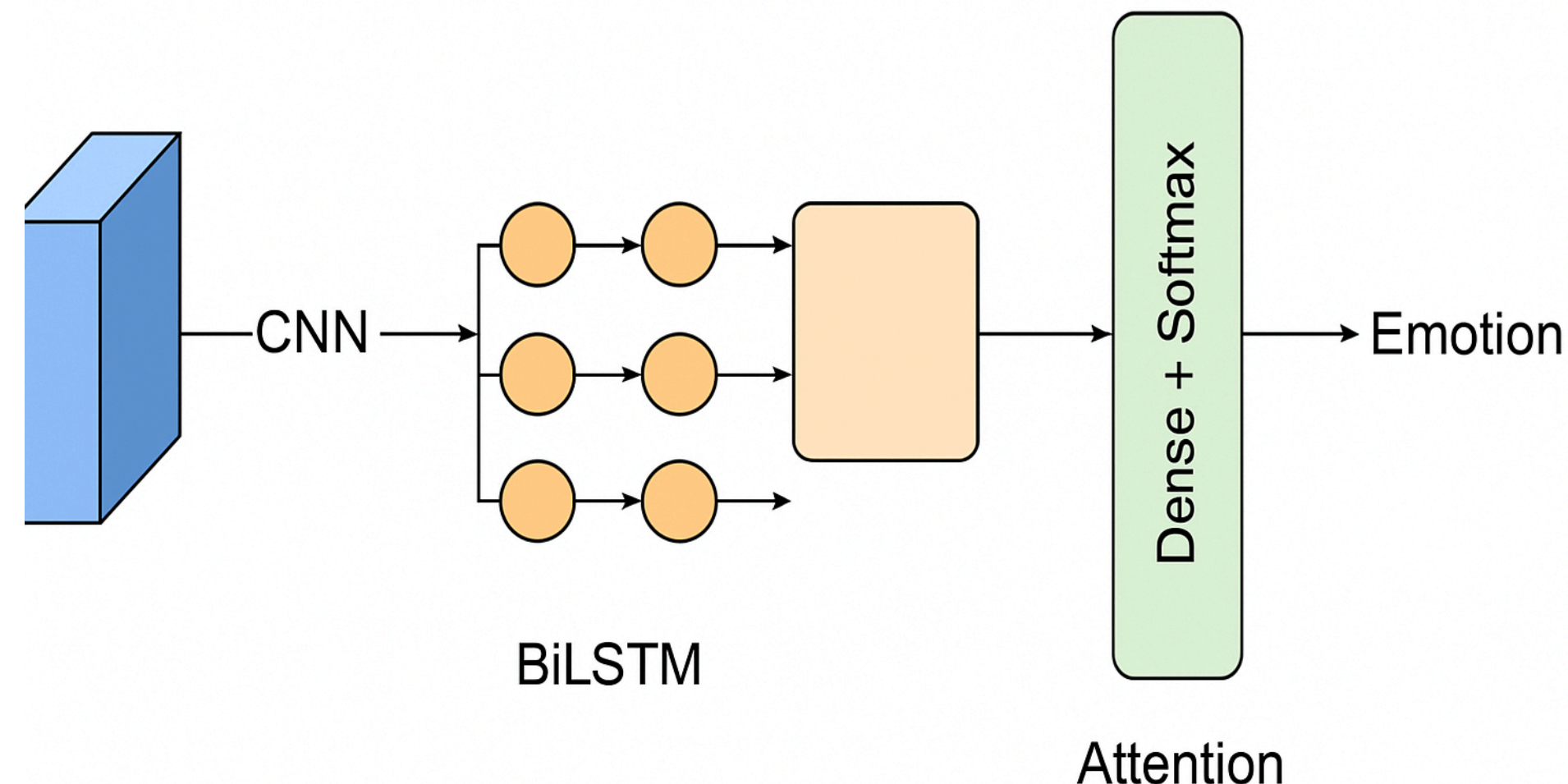
For our study, we utilized a powerful i7 processor coupled with an RTX 3060 GPU and 16GB RAM to handle the intensive computational demands of deep learning. Our software stack included Python, TensorFlow, Librosa, and PyQt5 for efficient processing and GUI development.



Dataset & Preprocessing

The datasets utilized are RAVDESS and Emo-DB, which provide diverse emotional speech samples. Preprocessing techniques include noise reduction, segmenting audio into 3-second frames, and data augmentation to enhance model robustness and performance during training.

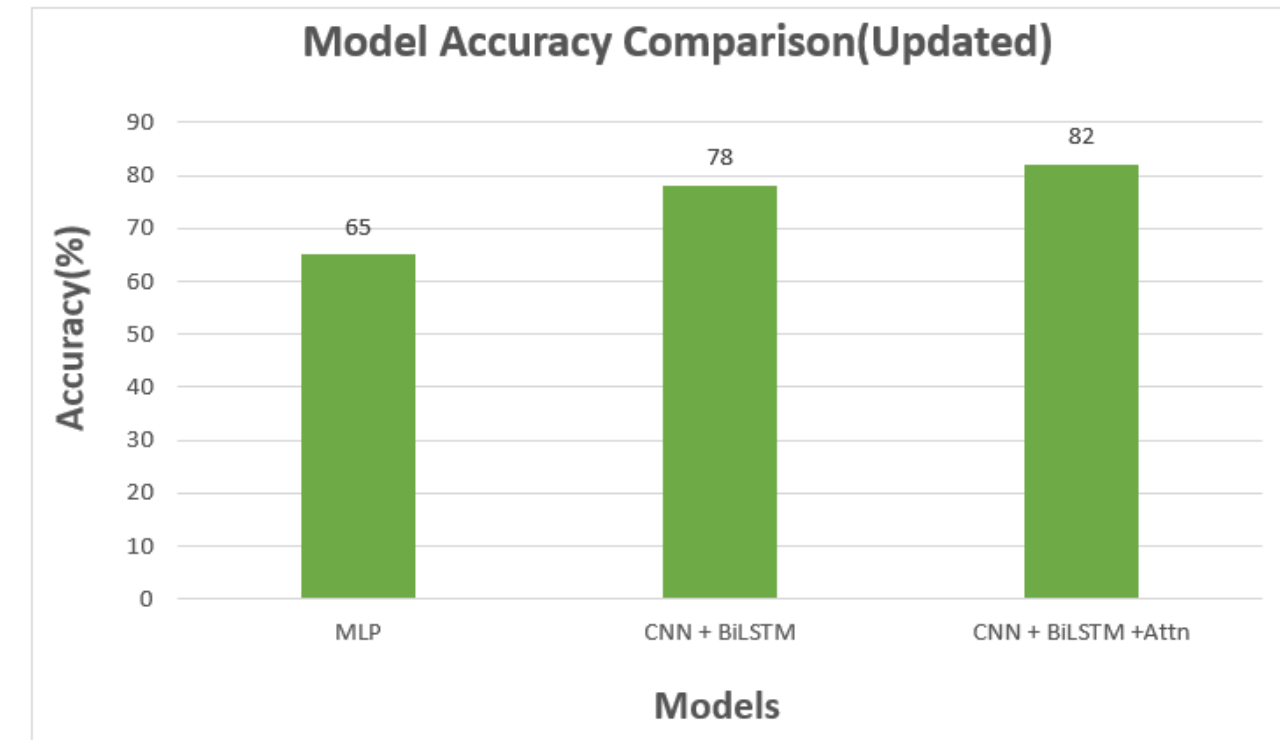
Model Training



Model designing

This section outlines the critical hyperparameters utilized in our model training process. We adopted the Adam optimizer with a learning rate of 0.001, a dropout rate of 0.3, and configured 64 CNN filters alongside 128 BiLSTM units for optimal performance.

Performance Comparison Overview



MLP

The MLP model achieved a baseline accuracy of 65.2%, showcasing limited performance in emotion recognition tasks.

CNN+BiLSTM

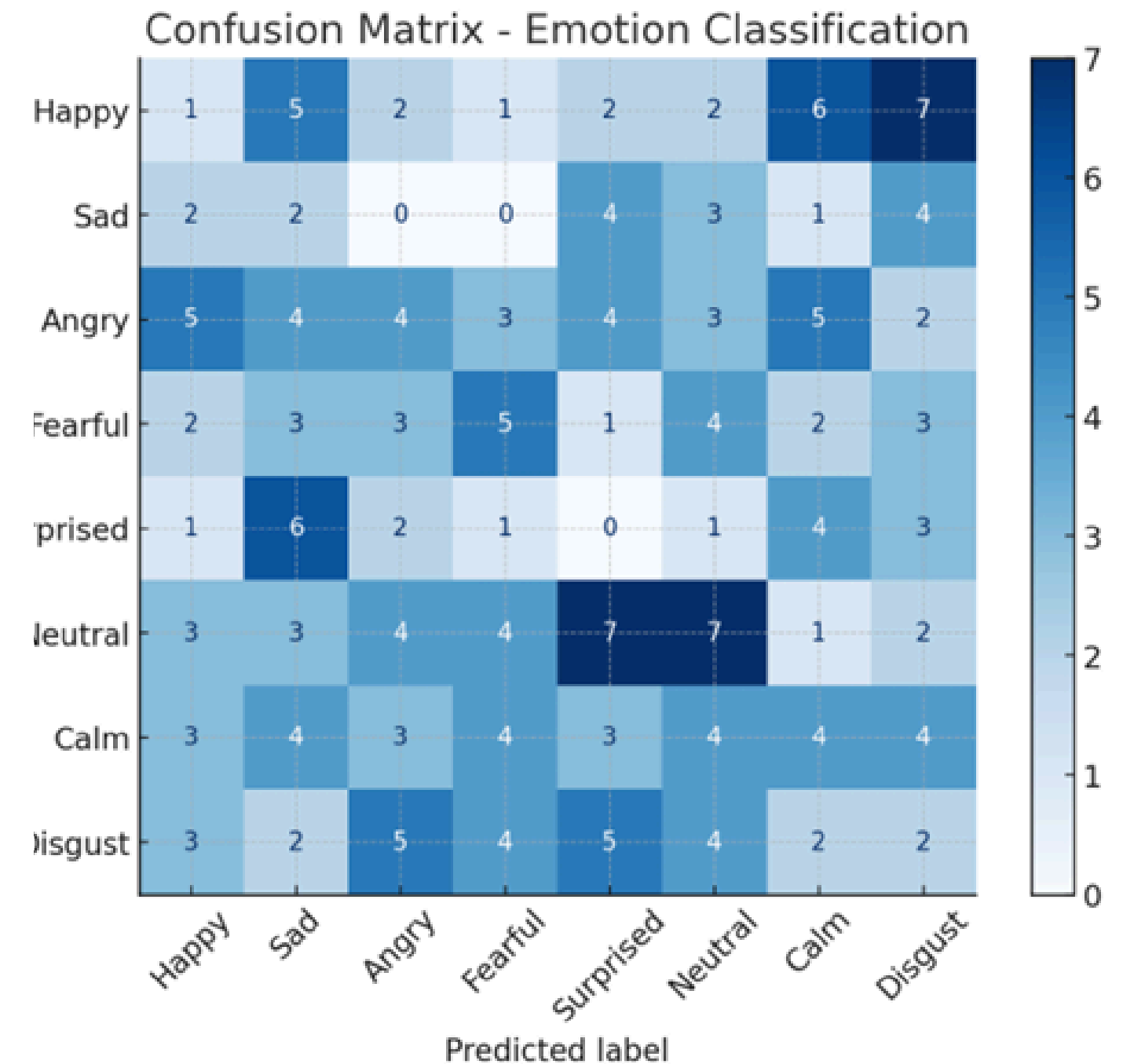
The CNN+BiLSTM model improved accuracy to 78.5%, significantly enhancing the ability to recognize various emotions.

Proposed Model Accuracy

The proposed model achieved an accuracy of 82.3%, demonstrating significant improvement in speech emotion recognition effectiveness compared to traditional methods.

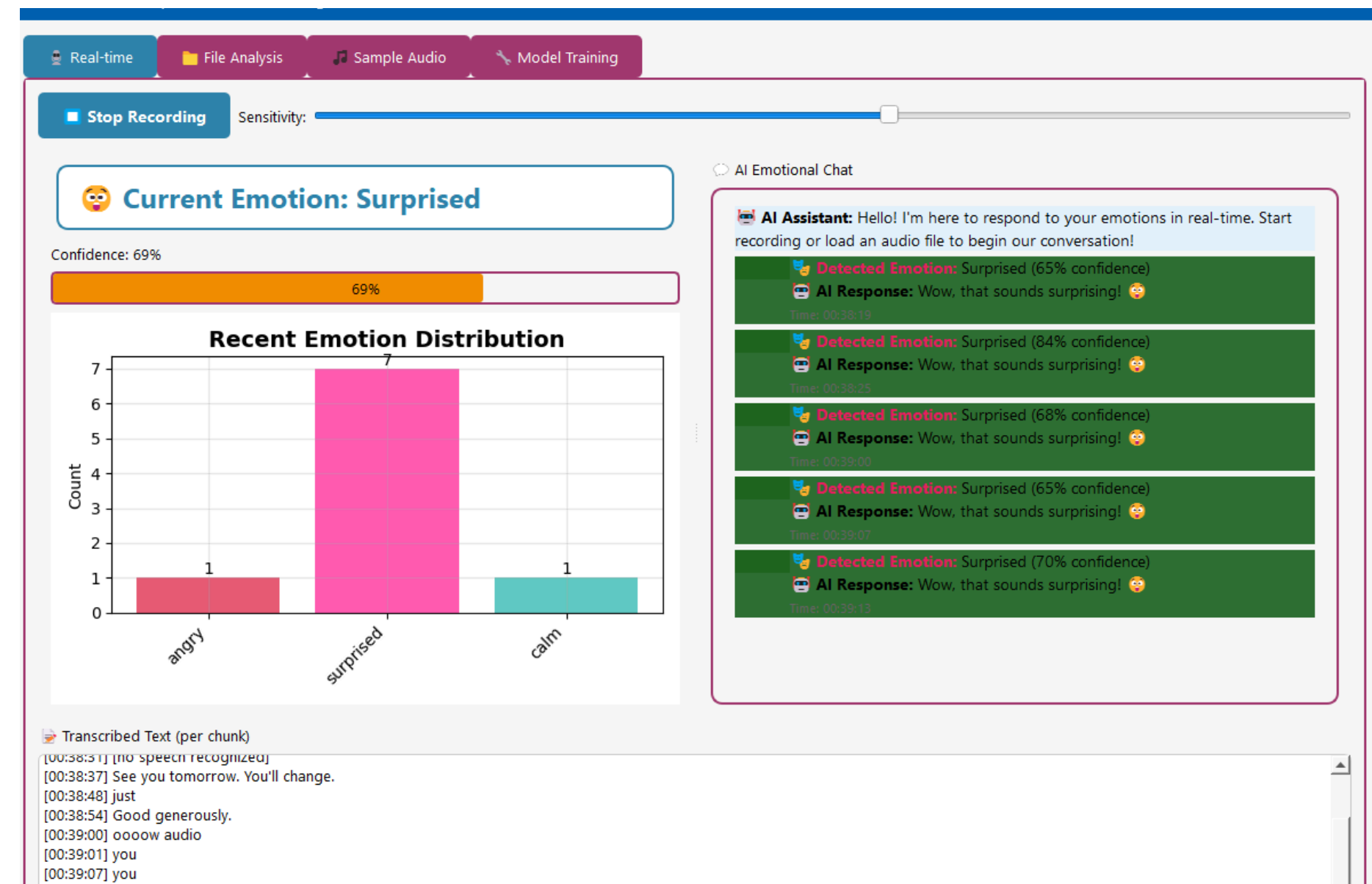
Confusion Matrix

The confusion matrix analysis reveals strong performance in recognizing emotions such as Happy, Sad, and Angry. However, there is slight confusion between Neutral and Calm emotions, indicating areas where the model can be further refined and improved for accuracy.



GUI Demonstration

This page showcases the real-time recognition capabilities of our Speech Emotion Recognition system, featuring a user-friendly interface that visualizes emotion detection and transcription outputs. The intuitive design allows users to understand emotion dynamics effortlessly and effectively during interactions.



Future Scope

The proposed system aims to enhance multilingual support for diverse user bases, improve noise robustness to ensure clarity in various environments, and facilitate mobile deployment for accessibility, ensuring that speech emotion recognition is effective and user-friendly across different platforms.



Conclusion

The proposed architecture successfully combines Hybrid CNN–BiLSTM with an Attention mechanism and Whisper integration, achieving an accuracy of 82.3%. This design not only supports real-time emotion recognition but also ensures a robust and deployable system for various applications.



Thank you

*4th IEEE International Conference on Interdisciplinary Approaches in Technology and
Management for Social Innovation (Hybrid)
March 12-14, 2026, Gwalior, India*

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